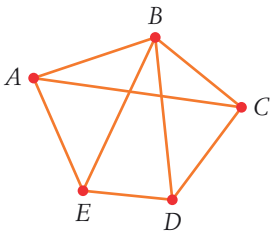
WORKSHEET

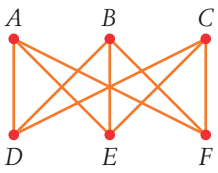
Chapter 5: Planar graphs

1 Label each of the following graphs as either planar or non-planar.

a

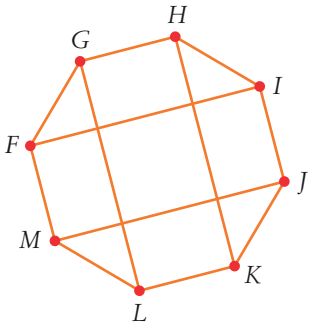


b

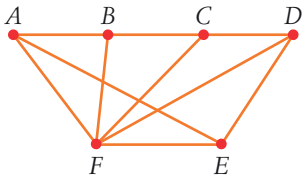


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c

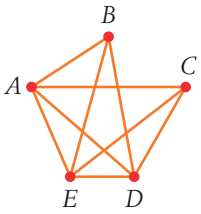


d

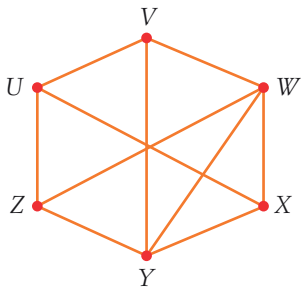


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e



f



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2 Redraw the planar graphs from Question **1** so that they have no intersecting edges.

3 Use Euler's formula to show that the graphs are planar.

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Answers

1 **a** Planar

b Non-planar

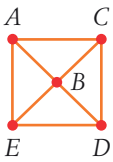
c Planar

d Planar

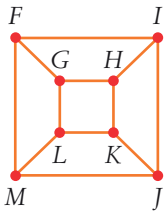
e Planar

f Non-planar

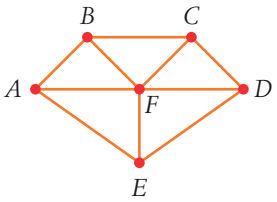
2 **a**



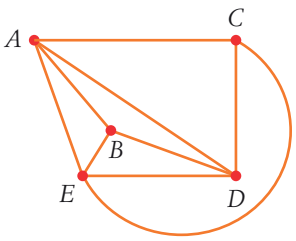
c



d



e



3 **a** $V = 5, E = 8, F = 5$

c $V = 8, E = 12, F = 6$

d $V = 6, E = 10, F = 6$

e $V = 5, E = 9, F = 6$

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